Traffic Engineering

METI

Lab 1 Report [EN]

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Group X

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1 Introduction

The objective of this laboratory assignment was to simulate and analyse the behaviour of Poisson arrival processes and compare the experimental results with their theoretical probability distributions.

The work developed so far includes:

- Simulation of independent Poisson arrival processes;
- Manual construction of histograms without using built-in histogram functions;
- Comparison between experimental and theoretical Poisson distributions;
- Analysis for different values of λ;
- Superposition of multiple independent Poisson processes.

The implementation was developed in Python using random number generation, numerical functions and graphical visualization tools.

2 Poisson Process Simulation

2.1 Theoretical Background

A Poisson process with parameter λ represents a random arrival process in which the number of events occurring in a unit time interval follows a Poisson distribution:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{1}$$

where:

- k = number of events in one interval;
- λ = average arrival rate.

The time between two consecutive arrivals follows an exponential distribution:

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0 \tag{2}$$

To generate exponentially distributed inter-arrival times, the inverse transform method was used:

$$\Delta t = \frac{-\ln(1-u)}{\lambda} \tag{3}$$

where u is uniformly distributed in $[0, 1)$.

2.2 Implementation Method

For each experiment:

1. Generate N inter-arrival times Δt;
2. Compute cumulative arrival instants;
3. Divide time into unit intervals;
4. Count the number of arrivals in each interval;
5. Build the histogram manually;
6. Compare experimental probabilities with the theoretical Poisson law.

The use of built-in histogram functions was intentionally avoided, as requested in the assignment statement.

2.3 Results for Different Values of λ

The simulator was executed for:

$$\lambda = \{0.5,\ 1,\ 5,\ 10,\ 50\}$$

For each value, the number of generated samples was increased proportionally in order to reduce statistical noise for large λ values.

$\lambda = 0.5$

For small values of λ, most intervals contain zero or one arrival. This behaviour matches the theoretical expectation, where $P(0)$ is dominant.

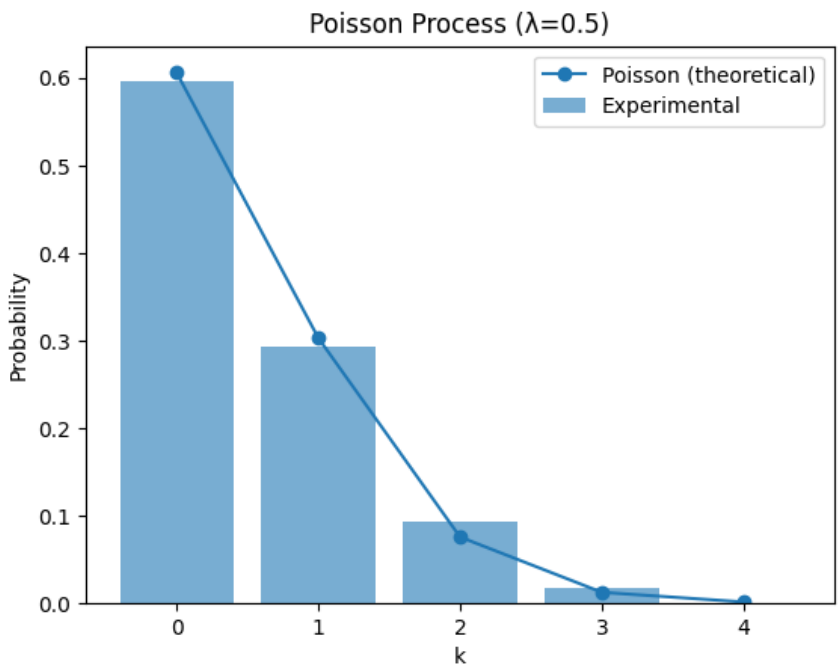

Figure 1: Experimental and theoretical distribution for $\lambda = 0.5$.

The probability mass remains concentrated at low values of k, although intervals with two or more arrivals become more frequent.

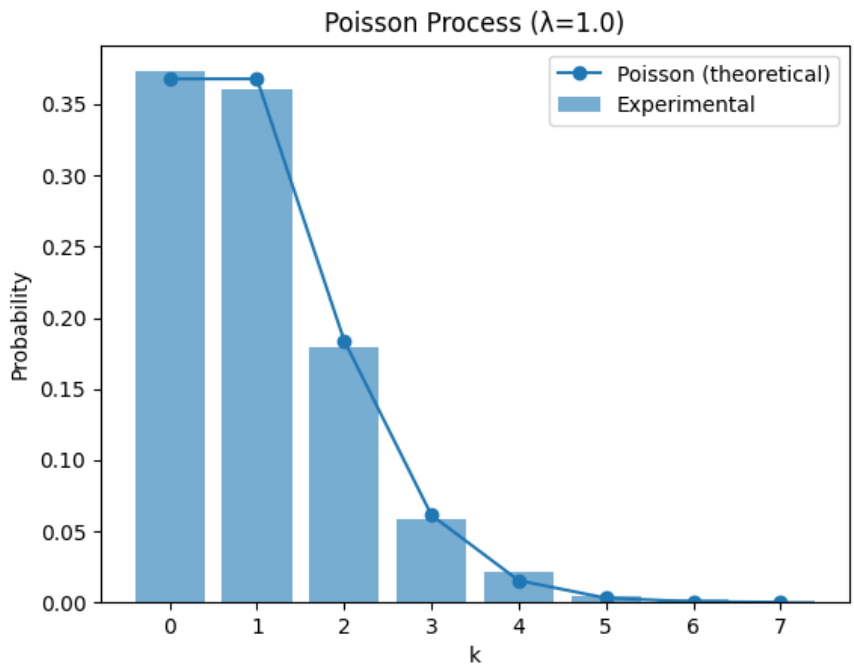

Figure 2: Experimental and theoretical distribution for $\lambda = 1$.

The distribution becomes wider and more symmetric around the mean value.

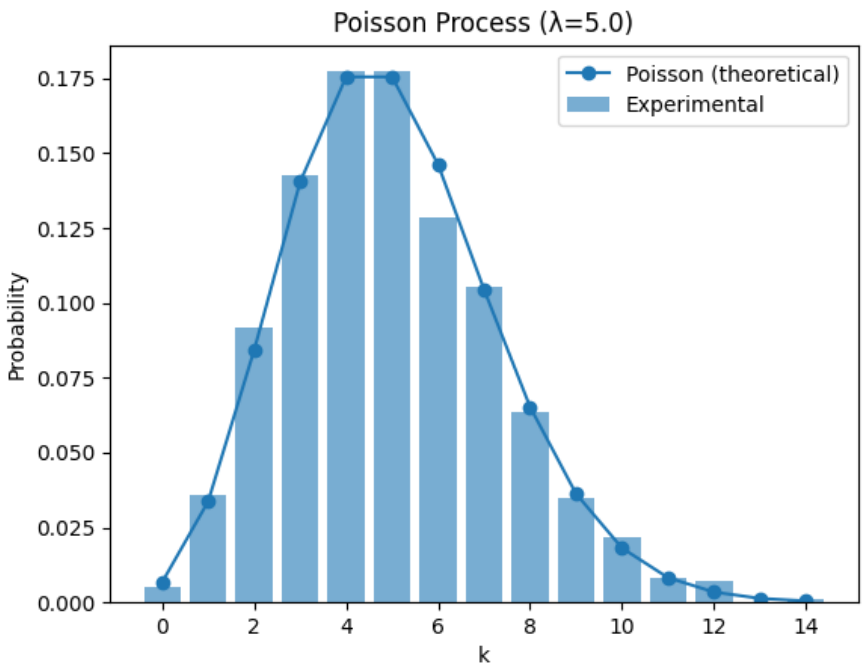

Figure 3: Experimental and theoretical distribution for $\lambda = 5$.

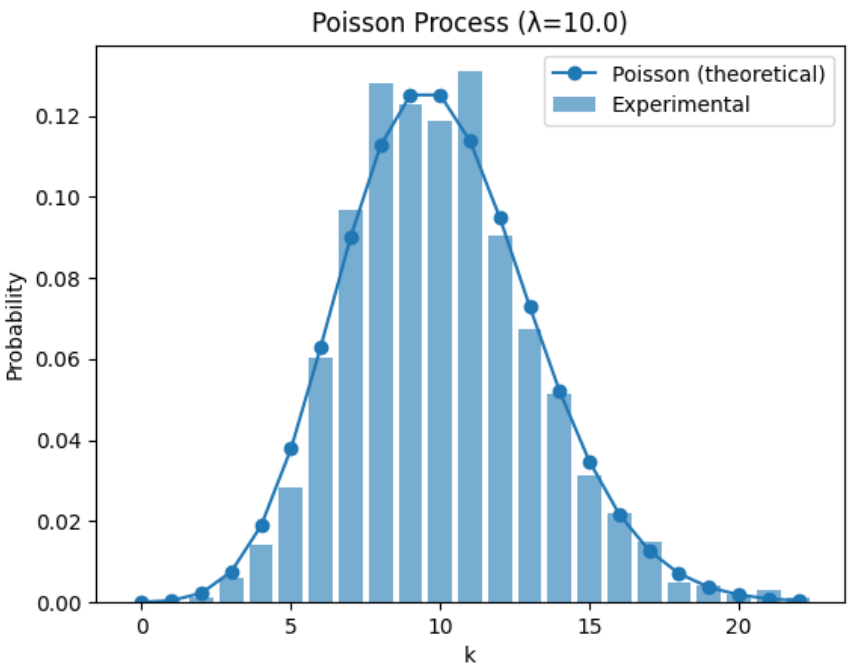

Figure 4: Experimental and theoretical distribution for $\lambda = 10$.

For very large arrival rates, the Poisson distribution becomes approximately bell-shaped due to convergence towards a normal distribution.

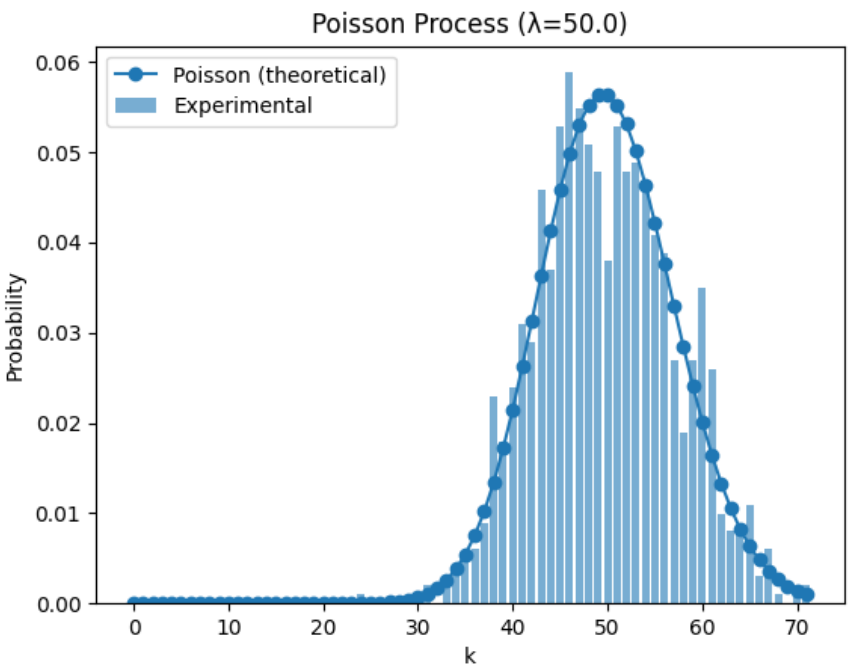

Figure 5: Experimental and theoretical distribution for $\lambda = 50$.

2.4 Discussion

The obtained results validate the simulator implementation. As the number of generated events increases, the experimental distribution converges to the theoretical model.

It was also observed that:

- Small λ values produce sparse arrival patterns;
- Larger λ values generate more regular and concentrated distributions;
- Increasing the sample size improves accuracy and reduces random fluctuations.

3 Superposition of Poisson Processes

3.1 Theoretical Background

A fundamental property of Poisson processes states that the superposition of independent Poisson processes is also a Poisson process whose rate is equal to the sum of the individual rates.

Given four independent processes:

$$\lambda_1 = 2, \quad \lambda_2 = 9, \quad \lambda_3 = 11, \quad \lambda_4 = 15$$

the resulting combined process has parameter:

$$\lambda_{total} = 2 + 9 + 11 + 15 = 37 \tag{4}$$

3.2 Implementation

Four independent processes were generated over the same observation interval. Their arrival instants were merged and sorted chronologically, producing a single combined arrival stream.

A histogram was then constructed using the same methodology described previously.

3.3 Results

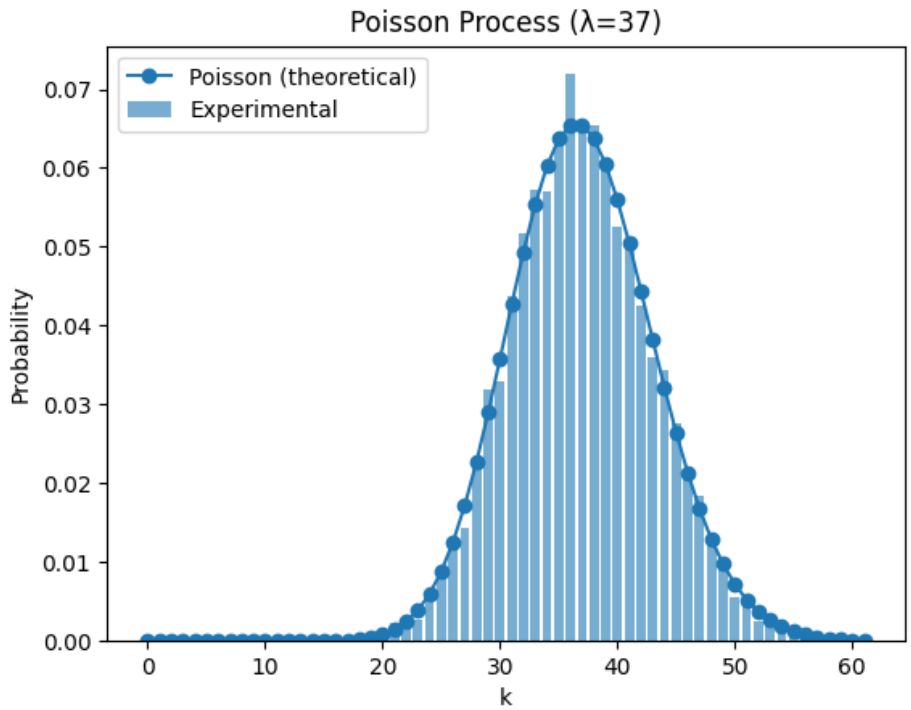

Figure 6: Experimental and theoretical distribution for the superposed process ($\lambda = 37$).

The experimental histogram is centered near the expected mean value and follows the theoretical Poisson distribution with good accuracy.

3.4 Discussion

This experiment confirms the closure property of Poisson processes under superposition. The sum of independent arrival streams preserves the Poisson nature of the process while increasing the global arrival rate.

Such behaviour is highly relevant in communication networks, where traffic generated by multiple independent users can often be modelled as a single aggregate Poisson flow.

4 Conclusion

The objectives defined up to Section 2.3 of the assignment were successfully completed. The developed simulator correctly reproduces:

- Individual Poisson arrival processes;
- Theoretical event-count distributions;
- Statistical convergence with larger sample sizes;
- Superposition of independent Poisson processes.